

Sales & Demand Forecasting for Businesses

1. Introduction

Sales forecasting helps organizations predict future demand using historical sales data. The objective of this project was to build a forecasting model capable of estimating future sales trends and supporting business decision-making.

2. Dataset Description

The dataset contains historical sales records including order dates and sales values. Historical sales data was used to identify trends and patterns for forecasting future sales.

3. Data Cleaning

Performed missing value checks, handled null records, converted date columns to datetime format, aggregated sales by date, and verified data consistency.

4. Feature Engineering

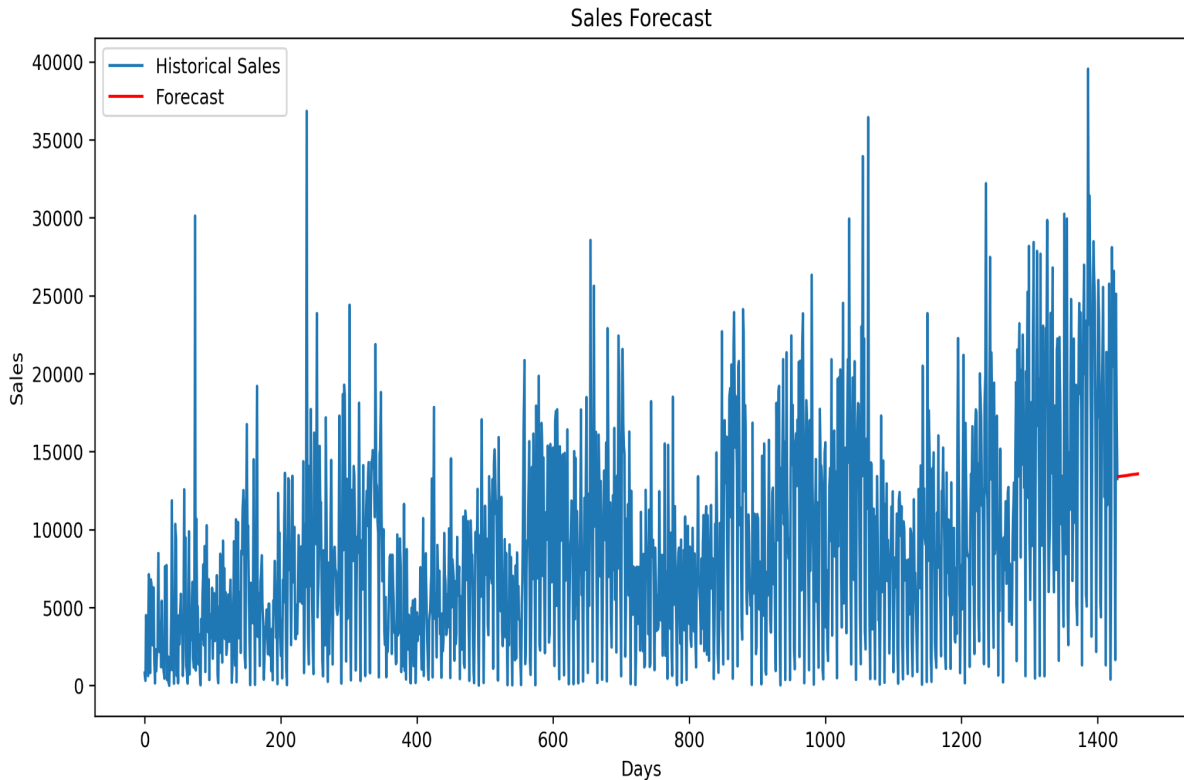
Created Year, Month, Day, and Day Number features from the order date to capture time-based sales patterns.

5. Forecasting Model

A Linear Regression model was used to forecast future sales. The model was trained using historical sales trends and then used to predict future demand.

6. Forecast Results

The forecasting model successfully generated future sales predictions based on historical sales patterns.



7. Business Insights

- Historical sales show an upward trend.
- Forecasted sales indicate stable growth.
- Inventory planning can be improved.
- Resource allocation can be optimized.
- Forecasting supports strategic decision-making.

8. Tools & Technologies Used

Python, Pandas, NumPy, Matplotlib, Scikit-Learn, Google Colab, GitHub.

9. Conclusion

This project successfully developed a sales forecasting model and generated useful business insights.

GitHub Repository

<https://github.com/devagirisivanagendrareddy9474-web/Sales-Demand-Forecasting>